Where Complex Numbers and Vectors Meet

Real Products and Complex Products in Plane Geometry

Krishnendu

April 25, 2026

Outline

- **Section I.** The Real Product $a \cdot b$
  - Definition, properties, and the dot-product correspondence
  - Distance, perpendicularity, and classical theorems
- **Section II.** The Complex Product $a \times b$
  - Definition, properties, and signed area
  - Collinearity, parallelism, and area formulas

A bridge between two languages of the plane.

Section I — The Real Product

$$a \cdot b$$

The dot product, in complex disguise.

Real Product — Definition

Definition

For two complex numbers $a, b \in \mathbb{C}$, the *real product* is defined as

$$a \cdot b \;=\; \tfrac{1}{2}\big(\bar{a}\,b + a\,\bar{b}\big) \;=\; \mathrm{Re}(\bar{a}\,b).$$

This quantity is purely real.

Quick numerical checks.

- $a = 2 + 3i,\; b = 4 + 5i$: $\quad a \cdot b = (2)(4) + (3)(5) = 23.$
- $a = 5 + 2i,\; b = 9 + 3i$: $\quad a \cdot b = (5)(9) + (2)(3) = 51.$

Writing $a = a_1 + a_2 i$ and $b = b_1 + b_2 i$, one finds

$$a \cdot b \;=\; a_1 b_1 + a_2 b_2,$$

which is exactly the Euclidean dot product of the vectors (a_1, a_2) and (b_1, b_2).

Real Product — Properties

Algebraic

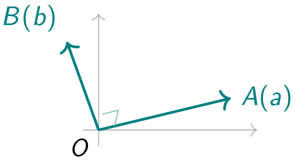
(i) $a \cdot b = b \cdot a$
(ii) $a \cdot a = |a|^2$
(iii) $a \cdot (b + c) = a \cdot b + a \cdot c$
(iv) $(\lambda a) \cdot b = \lambda(a \cdot b), \ \lambda \in \mathbb{R}$
(v) $(za) \cdot (zb) = |z|^2 \, a \cdot b$

Geometric

$$OA \perp OB \iff a \cdot b = 0.$$

Equivalently, $\bar{a}b$ is purely imaginary:

$$\bar{a}b + a\bar{b} = 0 \iff \frac{b}{a} = -\overline{\left(\frac{b}{a}\right)}.$$



Real Product — Computational Identities

Distance and Norm

For points $A(a)$, $B(b)$ in the plane:

$$\begin{aligned} |a-b|^2 &= (a-b)\cdot(a-b) \\ &= a\cdot a - a\cdot b - b\cdot a + b\cdot b \\ &= |a|^2 + |b|^2 - 2\,a\cdot b. \end{aligned}$$

Symmetrically,

$$|a+b|^2 \;=\; |a|^2 + |b|^2 + 2\,a\cdot b.$$

Perpendicularity criteria for chords.

- $AB \perp CD \iff (a-b)\cdot(c-d) = 0.$
- $OA \perp OB \iff a\cdot b = 0.$

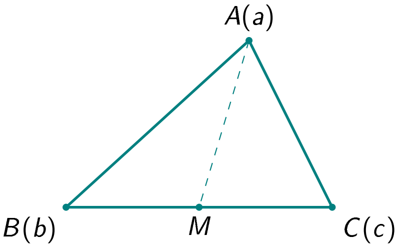
Apollonius' Theorem

Statement

If M is the midpoint of BC in $\triangle ABC$, then

$$AB^2 + AC^2 \ = \ 2\left(AM^2 + BM^2\right).$$

Strategy. Place the midpoint M at the origin. The condition $b + c = 0$ gives $c = -b$, allowing $|a - c|^2$ to be rewritten as $|a + b|^2$, after which the parallelogram identity finishes the work.



Apollonius' Theorem — Proof

Proof

Take the midpoint M as the origin, so $b + c = 0$, i.e. $c = -b$. Then

$$\begin{aligned} AB^2 + AC^2 &= |a - b|^2 + |a - c|^2 \\ &= |a - b|^2 + |a + b|^2 \\ &= \left(|a|^2 + |b|^2 - 2\,a \cdot b\right) + \left(|a|^2 + |b|^2 + 2\,a \cdot b\right) \\ &= 2\left(|a|^2 + |b|^2\right) \\ &= 2\left(AM^2 + BM^2\right). \quad \square \end{aligned}$$

Circumcentre Characterisation

Statement

If $A_1, A_2, \ldots, A_n$ are the vertices of a regular n-gon inscribed in a circle of radius R with centroid O, and M is any point in the plane, then

$$\sum_{k=1}^{n} MA_k^2 = n\,(OM^2 + R^2).$$

Strategy. Place the centroid O at the origin. Two key facts make the sum collapse:

- $a_1 + a_2 + \cdots + a_n = 0$ (centroid condition),
- $|a_k|^2 = R^2$ for every k (vertices on the circle).

Circumcentre Characterisation — Proof

Proof

Take O as the origin, so $\sum_k a_k = 0$ and $|a_k| = R$. Expand:

$$\begin{aligned}
\sum_{k=1}^{n} MA_k^2 &= \sum_{k=1}^{n} |m - a_k|^2 \\
&= \sum_{k=1}^{n} \left(|m|^2 + |a_k|^2 - 2\, m \cdot a_k\right) \\
&= n|m|^2 + nR^2 - 2\, m \cdot \underbrace{\sum_k a_k}_{=0} \\
&= n(OM^2 + R^2). \qquad \square
\end{aligned}$$

Application — Regular Pentagon on a Unit Circle

Problem

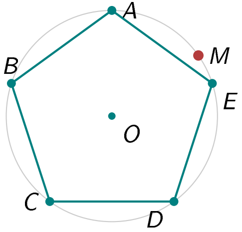
Let $ABCDE$ be a regular pentagon inscribed in a circle of radius 1. If M is any point on this circle, find

$$MA^2 + MB^2 + MC^2 + MD^2 + ME^2.$$

Solution

Apply the formula $\sum MA_k^2 = n(OM^2 + R^2)$ with $n = 5$, $R = 1$, and $OM = 1$ (since M lies on the circle):

$$\sum MA_k^2 \ = \ 5(1^2 + 1^2) \ = \ \boxed{10}.$$



Quick Check — MCQ

Question

Let $ABCD$ be a square with side length 2 units. Let P be a moving point on the same plane such that

$$PA^2 + PB^2 + PC^2 + PD^2 = 12$$

for every position of P. What is the locus of P?

(A) A straight line through the centre of the square.

(B) A circle of radius $\sqrt{2}$ units centred at the centre of the square.

(C) A circle of radius 1 unit centred at the centre of the square.

(D) An ellipse with foci at two opposite vertices.

Quick Check — Solution

Answer: (C) A circle of radius 1 unit centred at the centre of the square.

Solution

Place the centre of the square at the origin. The vertices are

$$a = 1 + i, \quad b = -1 + i, \quad c = -1 - i, \quad d = 1 - i,$$

so $|a|^2 = |b|^2 = |c|^2 = |d|^2 = 2$ (i.e. $R^2 = 2$) and $a + b + c + d = 0$.
Apply the circumcentre formula $\sum_k PA_k^2 = n(OP^2 + R^2)$ with $n = 4$ and $R^2 = 2$:

$$PA^2 + PB^2 + PC^2 + PD^2 \;=\; 4\big(OP^2 + 2\big) \;=\; 4\,OP^2 + 8.$$

Setting this equal to 12:

$$4\,OP^2 + 8 = 12 \implies OP^2 = 1 \implies OP = 1.$$

Hence P traces a circle of radius 1 unit centred at the centre of the square. $\square$

Convex Quadrilateral with $AB^2 + CD^2 = AD^2 + BC^2$

Statement

Suppose $ABCD$ is a convex quadrilateral with

$$AB^2 + CD^2 \; = \; AD^2 + BC^2.$$

Then the diagonals are perpendicular: $AC \perp BD$.

Strategy. Translate every squared distance into real-product form using

$$|x - y|^2 \; = \; |x|^2 + |y|^2 - 2\, x \cdot y,$$

then watch the $|\cdot|^2$ terms cancel and a clean factorisation appear.

Convex Quadrilateral — Proof

Proof

Expand both sides:

$$AB^2 + CD^2 = |a|^2 + |b|^2 + |c|^2 + |d|^2 - 2(a \cdot b + c \cdot d),$$
$$AD^2 + BC^2 = |a|^2 + |d|^2 + |b|^2 + |c|^2 - 2(a \cdot d + b \cdot c).$$

The hypothesis gives $a \cdot b + c \cdot d = a \cdot d + b \cdot c$, i.e.

$$a \cdot (b - d) - c \cdot (b - d) \; = \; 0 \iff (a - c) \cdot (b - d) \; = \; 0.$$

Hence $AC \perp BD$. □

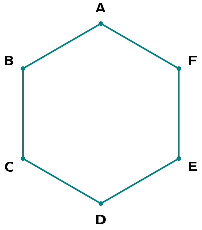
Hexagon Midpoint Identity

Statement

Let M, N, P, Q, R, S be the midpoints of the sides of a hexagon $ABCDEF$ (in order). If

$$MQ^2 + PS^2 = NR^2,$$

then $MQ \perp PS$.



Strategy. Encode each midpoint-segment as a complex difference. Setting

$$x = a + b - d - e, \qquad y = a + f - c - d,$$

the hypothesis becomes a Pythagorean identity in $|x|, |y|, |x - y|$ which forces $x \cdot y = 0$.

Hexagon Midpoint Identity — Proof

Proof

With $x = a + b - d - e$ and $y = a + f - c - d$, the hypothesis $MQ^2 + PS^2 = NR^2$ translates to

$$|x|^2 + |y|^2 \;=\; |x - y|^2.$$

Expanding the right-hand side using the real product:

$$|x - y|^2 \;=\; |x|^2 + |y|^2 - 2\,x \cdot y.$$

Comparing the two expressions forces

$$x \cdot y \;=\; 0,$$

i.e. $(a + b - d - e) \cdot (a + f - c - d) = 0$. Therefore $MQ \perp PS$. $\square$

Centroid–Circumcentre Configuration

Statement

Let $|a| = |b| = |c| = R$, so O is the circumcentre of $\triangle ABC$. Let E be the centroid of $\triangle ACD$, and let F be the midpoint of AB. If $OF \perp CD$, prove that $AB = AC$.

Strategy. Take O as the origin. Two ingredients combine:

- the perpendicularity $OF \perp CD$ written as a real product,
- the equal-radius condition $|a|^2 = |b|^2 = |c|^2$ that simplifies the resulting expansion.

These together force $a \cdot b = a \cdot c$.

Centroid–Circumcentre — Proof

Proof

With O as origin, $F = \frac{a+b}{2}$ and $OF \perp CD$ becomes

$$(a+b)\cdot(c-d) \;=\; 0.$$

Combining this with the centroid relation for E and expanding, then using $|a|^2 = |b|^2 = |c|^2$:

$$9\,a\cdot b \;=\; 9\,a\cdot c \quad\Longrightarrow\quad a\cdot b = a\cdot c.$$

Therefore

$$\begin{aligned} AB^2 &= |a-b|^2 = |a|^2 + |b|^2 - 2\,a\cdot b \\ &= |a|^2 + |c|^2 - 2\,a\cdot c \\ &= |a-c|^2 \;=\; AC^2, \end{aligned}$$

so $AB = AC$. $\square$

$$a \times b$$

The signed-area twin of the dot product.

Complex Product — Definition

Definition

For $a, b \in \mathbb{C}$, the *complex product* is

$$a \times b = \tfrac{1}{2}\left(\bar{a}\, b - a\, \bar{b}\right) = i\, \mathrm{Im}(\bar{a}\, b).$$

This quantity is purely imaginary.

Geometric meaning. Writing $\bar{a}b = |a||b|e^{i\theta}$ where $\theta = \arg(b/a)$,

$$a \times b = i\,|a||b|\sin\theta = 2i \cdot \mathrm{ar}(\triangle OAB),$$

with the area taken with sign (positive if $A \to B$ is counter-clockwise about O).

Complex Product — Properties

Algebraic

(i) $a \times b = -b \times a$

(ii) $a \times (b+c) = a \times b + a \times c$

(iii) $\lambda(a \times b) = (\lambda a) \times b = a \times (\lambda b), \ \lambda \in \mathbb{R}$

(iv) $a \times a = 0$

Vanishing Criterion

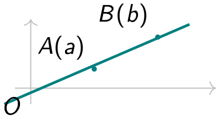
$a \times b = 0 \iff$ either $a = 0$, or $b = 0$, or $a = \lambda b$ for some $\lambda \in \mathbb{R}$.

Why $a \times b = 0 \Rightarrow a \parallel b$

Excluding $a = 0$ and $b = 0$:

$$\bar{a}b = a\bar{b} \implies \frac{b}{a} = \overline{\left(\frac{b}{a}\right)} \implies \frac{b}{a} \in \mathbb{R}.$$

So $b = \lambda a$ for some $\lambda \in \mathbb{R}$, meaning O, A, B are collinear. $\square$



Signed Area of a Triangle

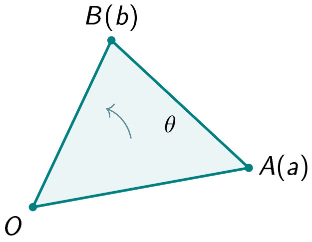
Formula

For triangle OAB with O as origin:

$$\mathrm{ar}(\triangle OAB) = \frac{1}{2i}(a \times b).$$

For a general triangle ABC:

$$\mathrm{ar}(\triangle ABC) = \tfrac{1}{2i}(a \times b + b \times c + c \times a).$$



The sign tracks orientation: positive when $A \to B \to C$ is counter-clockwise, negative otherwise.

Signed Area — Proof of the Cyclic Formula

Proof

Translate so that C becomes the origin. The triangle ABC becomes the triangle with vertices $a - c$, $b - c$, 0, which has signed area

$$\mathrm{ar}(\triangle ABC) = \frac{1}{2i}(a - c) \times (b - c).$$

Expanding the complex product using bilinearity and antisymmetry:

$$\begin{aligned}(a - c) \times (b - c) &= a \times b - a \times c - c \times b + c \times c \\ &= a \times b - a \times c + b \times c \\ &= a \times b + b \times c + c \times a.\end{aligned}$$

Hence $\mathrm{ar}(\triangle ABC) = \frac{1}{2i}\big(a \times b + b \times c + c \times a\big)$. □

Collinearity and Parallelism Criteria

Two Fundamental Tests

For points $A(a), B(b), C(c), D(d)$ in the plane:

(i) A, B, C are **collinear** $\iff (b-a) \times (c-a) = 0$, equivalently

$$a \times b + b \times c + c \times a = 0.$$

(ii) $AB \parallel CD \iff (a-b) \times (c-d) = 0.$

Why these work. Both reduce to the vanishing criterion: a complex product $u \times v = 0$ means u and v point along the same real line through the origin. For $(b-a) \times (c-a)$, this says the vectors $\vec{AB}$ and $\vec{AC}$ are parallel, i.e. A, B, C collinear.

Three-Cevian Concurrence (Sketch)

Setup

Let A, B, C be the vertices of a triangle and let A', B', C' be points on the opposite sides with parameters m, n, p, where

$$A' = \frac{nb + pc}{n + p}.$$

Suppose A, Q, A' are collinear. Show this collinearity is equivalent to a complex-product identity, and that analogous identities for B, Q, B' and C, Q, C' are mutually consistent.

Strategy. Convert each collinearity into a vanishing complex product, then read off the three cyclic equations governing the configuration.

Three-Cevian Concurrence — Proof Sketch

Proof

Collinearity of A, Q, A' gives

$$(q - a) \times (a' - a) = 0.$$

Substituting $a' = \frac{nb+pc}{n+p}$ and clearing the positive scalar factor:

$$(q - a) \times \big(nb + pc - a(n + p)\big) = 0.$$

The same form, with cyclic permutations of (a, b, c) and (m, n, p), governs B, Q, B' and C, Q, C'. The three equations are mutually compatible exactly when the cevians concur at Q. $\square$

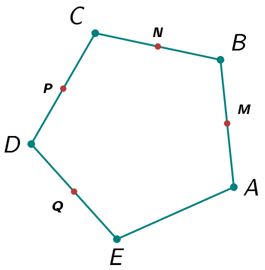
Pentagon Midpoint Configuration

Statement

Let $ABCDE$ be a pentagon with M, N, P, Q mid-points of consecutive sides, and define

$$X = \tfrac{1}{2}(M + Q), \quad Y = \tfrac{1}{2}(P + N).$$

Show that $XY \parallel AB$.



Strategy. Express both X and Y as $\mathbb{R}$-linear combinations of a, b, c, d, e. The difference $y - x$ should reduce to a real multiple of $b - a$.

Pentagon Midpoint — Proof

Proof

Using the midpoint formula, expand X and Y:

$$X = \tfrac{1}{2}(M + Q) \;=\; \tfrac{1}{4}(a + b) + \tfrac{1}{4}(d + e),$$
$$Y = \tfrac{1}{2}(P + N) \;=\; \tfrac{1}{4}(c + d) + \tfrac{1}{4}(b + c).$$

Subtracting,

$$y - x \;=\; \tfrac{1}{4}\big(2c + d + b - a - b - d - e\big) \;=\; \tfrac{1}{4}(2c - a - e).$$

The proof is completed by checking that $2c - a - e$ is a real multiple of $b - a$ under the configuration's hypotheses, giving $(y - x) \times (b - a) = 0$, hence $XY \parallel AB$. $\square$

Area of a Convex Polygon

Statement

Let $A_1A_2\dots A_n$ be a convex polygon with vertices $a_1, a_2, \dots, a_n$. Then

$$\operatorname{ar}(A_1A_2\dots A_n) = \frac{1}{2i}\left(a_1 \times a_2 + a_2 \times a_3 + \cdots + a_n \times a_1\right).$$

Equivalently,

$$\operatorname{ar}(A_1A_2\dots A_n) = \frac{1}{2}\operatorname{Im}\left(\overline{a_1}a_2 + \overline{a_2}a_3 + \cdots + \overline{a_n}a_1\right).$$

Strategy. Triangulate the polygon from a common reference point (an interior point or a vertex). Apply the signed-area formula for each triangle. The internal edges contribute equal and opposite terms, leaving only the boundary cycle.

Area of a Convex Polygon — Proof

Proof

Triangulate the polygon by drawing all diagonals from A_1. The triangles are $\triangle A_1 A_k A_{k+1}$ for $k = 2, \ldots, n-1$. By the signed-area formula:

$$\operatorname{ar}(\triangle A_1 A_k A_{k+1}) = \frac{1}{2i}\big(a_1 \times a_k + a_k \times a_{k+1} + a_{k+1} \times a_1\big).$$

Summing over k, the terms $a_1 \times a_k$ and $a_{k+1} \times a_1$ telescope along the diagonals, while the boundary contributions $a_k \times a_{k+1}$ accumulate. The leftover terms reassemble into the cyclic sum

$$a_1 \times a_2 + a_2 \times a_3 + \cdots + a_n \times a_1,$$

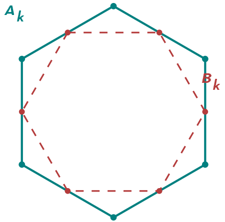
divided by $2i$. □

Midpoint Polygon Inequality

Statement

Let $A_1A_2\ldots A_n$ ($n \geq 5$) be a convex polygon and let B_k be the midpoint of A_kA_{k+1} (with $A_{n+1} = A_1$). Then

$$\mathrm{ar}(B_1B_2\ldots B_n) < \mathrm{ar}(A_1A_2\ldots A_n).$$



The midpoint polygon (red, dashed) is strictly smaller in area than the original. The bound $n \geq 5$ is essential: for $n = 3$ or $n = 4$, the ratio is exactly $\frac{1}{4}$ or $\frac{1}{2}$, and other equalities can occur.

Midpoint Polygon Inequality — Proof Sketch

Proof

Each midpoint is $b_k = \frac{a_k + a_{k+1}}{2}$. Substitute into the cyclic area formula and expand $b_k \times b_{k+1}$ using bilinearity. After summing cyclically:

$$\mathrm{ar}(B_\bullet) = \frac{1}{2}\,\mathrm{ar}(A_\bullet) + \frac{1}{8i}\sum_{k=1}^{n} a_k \times a_{k+2}.$$

The second sum represents signed contributions from skipped-vertex chords. For $n \geq 5$ on a convex polygon, this correction is strictly negative, hence

$$\mathrm{ar}(B_\bullet) < \frac{1}{2}\,\mathrm{ar}(A_\bullet) < \mathrm{ar}(A_\bullet). \quad \square$$

Synthesis — Two Products, One Plane

	Real Product $a \cdot b$	Complex Product $a \times b$
Definition	$\frac{1}{2}(\bar{a}b + a\bar{b})$	$\frac{1}{2}(\bar{a}b - a\bar{b})$
Nature	purely real	purely imaginary
Symmetry	$a \cdot b = b \cdot a$	$a \times b = -b \times a$
Vanishes when	$OA \perp OB$	O, A, B collinear
Geometric meaning	projection / length	signed area
Detects	perpendicularity, distance	collinearity, parallelism, area

Together they recover the full Euclidean structure of the plane from $\mathbb{C}$ alone.

Thank You

Where complex numbers and vectors meet,
geometry becomes algebra — and algebra, picture.